



Skole

How the system works

A plain-language guide to the school management system: who uses it, what each person can do, how the data fits together, and how it all runs.

Live site: <https://school-management-system-b0a2a.web.app>

Source code: <https://github.com/Lianhahaha/School-Management-System>

Demo sign-in: admin@school.test, teacher1@school.test or student1@school.test, password Password123!

What is in this guide

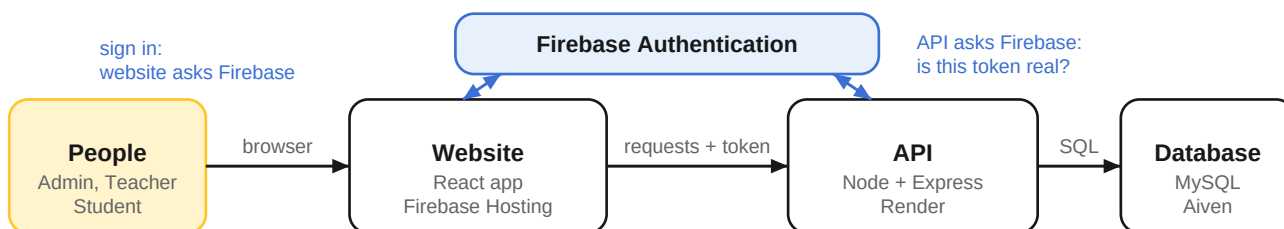
Part	What it answers
1. The big picture	What Skole is and which pieces it is made of
2. People and authority	Admin, teacher and student: who can do what, and who is above whom
3. Accounts and sign-in	How accounts are created, how login works, passwords, deactivating
4. How a school is organised	Classes, enrollments, timetables, attendance, grades
5. A day in the life	Step-by-step walk-throughs for each role
6. Rules the system enforces	The checks that keep the data correct
7. Dashboards and everyday features	What each person sees first, search, filters, themes
8. Under the hood	The technology, how a request travels, where the code lives
9. Hosting, data and limits	Where everything runs, what "real time" means, the free-tier limits
10. Questions and glossary	Short answers to common questions and plain meanings of the terms

1. The big picture

Skole is a web application for running a school. It is used in a normal browser on a phone, tablet or computer: there is nothing to install. Three kinds of people sign in, and each one sees only the parts that belong to them.

It is built from four pieces that each do one job:

Piece	Job	Where it runs
Website	The screens you see and click. Built with React.	Firebase Hosting (free)
API	The "back office". It checks who you are, enforces the rules and reads and writes data.	Render (free)
Database	Stores every school record: people, classes, attendance, grades, announcements.	Aiven MySQL (free)
Sign-in service	Knows your email and password and proves who you are. Stores no school data.	Firebase Authentication (free)



Every school record lives in MySQL. Firebase only knows who you are (email + password).

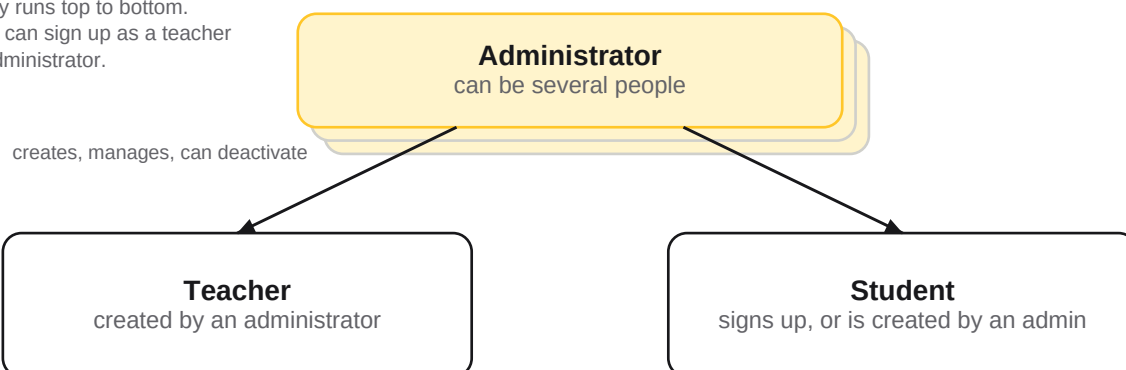
Why split it this way

The website never touches the database. Every action goes through the API, which checks who you are and what you may do first, so editing a web address cannot reveal anyone else's data.

2. People and authority

Skole has exactly three roles. A role is chosen when the account is created and it never changes afterwards.

Authority runs top to bottom.
Nobody can sign up as a teacher or an administrator.



The administrator

The administrator runs the school inside the system and is the highest authority. An administrator can see everything and change almost everything: accounts, students, teachers, subjects, classes, who teaches what, enrollments, timetables, attendance, grades and announcements.

There can be several administrators. The demo data has two (Amara Johnson and Noah Bennett), so you can safely deactivate one and still have the other. A school with a principal and a registrar would normally give both an admin account.

How admin accounts are made. Nobody can sign up as an administrator. An existing administrator creates the new one under Users, Create user, role Administrator. The very first administrator of a brand-new installation comes from the setup step (the seed command), which creates the two demo administrators.

The administrator is powerful, so the system protects itself

You cannot deactivate or delete your own account (it prevents locking yourself out).

The last active administrator can never be deactivated or deleted, even if two administrators try at the same moment.

An account that already has school records cannot be deleted, only deactivated, so history is never erased.

Role and email cannot be edited after creation. A wrong role is fixed by deleting the mistaken account (while it is still unused) and creating a new one.

The teacher

A teacher works with the classes they have been assigned. They mark attendance, create assessments and enter grades for *their own* lessons, see their timetable, post announcements to their classes, and read their students' information. They cannot create accounts, change classes or subjects, assign teachers, or touch another teacher's lessons.

The student

A student sees their own world: their class and subjects, timetable, attendance record, grades and the announcements meant for them. Everything is read-only except their own contact details (phone, address, guardian). A student can never see another student's records.

Who can do what

"Own" means only within their own scope: a teacher's own lessons, a student's own records.

Capability	Administrator	Teacher	Student
Create and deactivate accounts	Yes	No	No
Edit own name, phone, contact details	Yes	Yes	Yes
See student profiles	All	Students in their classes	Only themselves
Edit student and teacher profiles	Yes	No	Own contact details only
Subjects and classes: view	Yes	Yes	Yes
Subjects, classes, teacher assignment, enrollments, timetable: change	Yes	No	No
Attendance: mark and correct	Yes	Own lessons	No
Attendance: view	All	Own lessons	Own record
Assessments and grades: create, enter, edit	Yes	Own lessons	No
Assessments and grades: view	All	Own lessons	Own grades
Announcements: post	Anyone, any class	To a class they teach	No
Announcements: edit and delete	Any	Only their own	No
Announcements: read	All	Those aimed at teachers or their classes	Those aimed at students or their class

3. Accounts and sign-in

How each kind of account is created

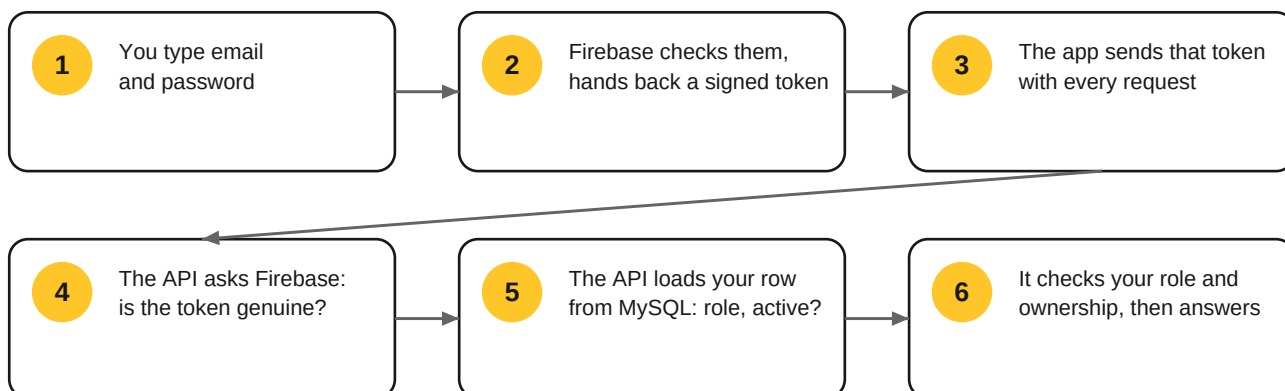
Role	How the account is made	Notes
Administrator	An administrator creates it: Users, Create user, role Administrator.	The first one comes from the setup (seed) command.
Teacher	An administrator creates it: Users, Create user, role Teacher. They enter name, email, a temporary password, and teacher details (employee number, hire date, department, qualification).	Teachers cannot sign up on their own. The administrator tells the teacher their email and temporary password.
Student	Either the student signs up on the public "Create account" page, or an administrator creates the account.	A self-registered student has no class yet. An administrator enrolls them, and only then do timetable, attendance and grades appear.

Why there is no "promote" button

A role decides which records an account owns: a student has enrollments and grades, a teacher has classes and a timetable. Turning one into the other would leave those records meaningless, so roles are fixed. If a teacher signed up as a student by mistake, the administrator deletes that unused account and creates a proper teacher account.

What happens when you sign in

Sign-in is shared between Firebase (which knows passwords) and the API (which knows roles and school data). The password never reaches the school database.



This check happens on **every single request**, not just at login. So if an administrator deactivates an account, that person is locked out on their very next click, even if they are already signed in.

Passwords and "Forgot password"

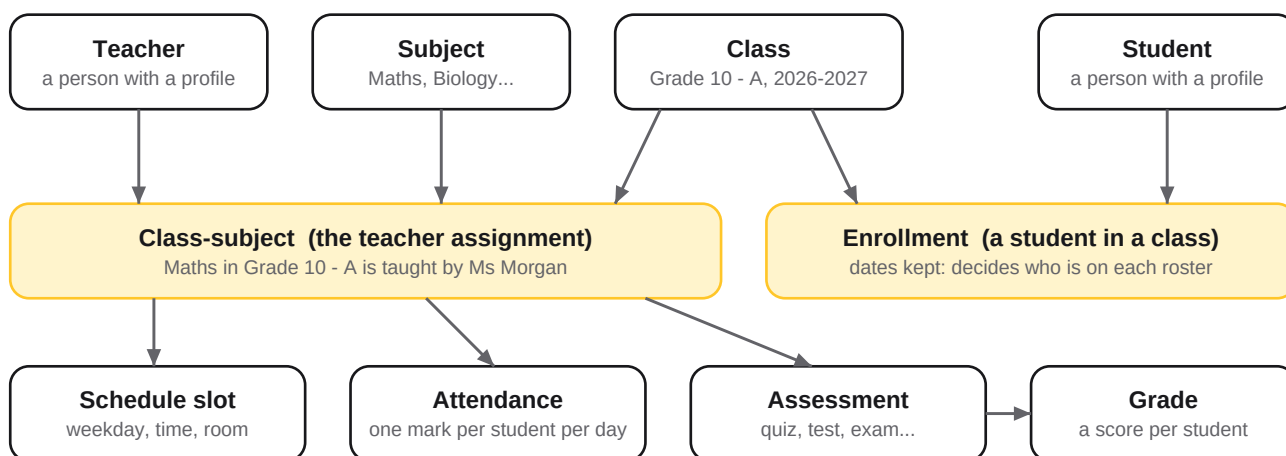
- Forgot password sends an **email with a reset link** (not a code). The link opens a Firebase page where the person types a new password, then they sign in again.
- The screen shows the same "check your email" message whether or not the email exists, so nobody can use it to discover who has an account.
- A signed-in person can also request the reset link from their **Profile** page.
- The demo accounts use fake addresses (for example admin@school.test), so they cannot receive email. To try the flow, register a student with a real email address.

Deactivating, deleting and what it does

Action	Effect
Deactivate an account	The person is blocked at once and their Firebase login is disabled. Nothing is deleted. Reactivating restores access. For a student , deactivating also closes their active enrollment as "withdrawn" (reactivating does not restore it).
Deactivate a teacher	Refused while the teacher still teaches subjects or is a homeroom teacher this school year. Reassign those first.
Delete an account	Only for a mistaken account with no school records at all. Removes the school record and the Firebase login, and frees the email. Refused (with a clear message) if any history exists, if it is your own account, or if it is the last administrator.

4. How a school is organised

Everything in Skole hangs off a few simple ideas. The diagram shows how they connect; the sections below explain each one.



Subjects and classes

A **subject** is something taught (Mathematics, Biology). A **class** is a group for one school year, with a name, a grade level (1 to 12) and an optional **homeroom teacher**, for example "Grade 10 - A, 2026-2027". The school year starts in **August**, so 3 October 2026 belongs to 2026-2027 and so does 10 May 2027.

A subject can be **retired** instead of deleted: it stops appearing in new assignments but old records stay. A subject or class that is in use cannot be deleted; the system says it is in use.

Teacher assignment (class-subject)

A **class-subject** pairs a class with a subject and the teacher who teaches it ("Mathematics in Grade 10 - A, taught by Ms Morgan"). It is the anchor for the timetable, attendance and grades, so a teacher's rights always trace back to being assigned here. Changing the teacher keeps all earlier attendance and grades with the subject.

Enrollment

An **enrollment** puts a student into a class, with a start date and, once finished, an end date and a reason: *active*, *transferred*, *completed* or *withdrawn*. A student can have **only one active enrollment**. Moving a student to another class ("transfer") closes the old enrollment and opens a new one in one step. Old enrollments are never overwritten, so a student's history, including leaving and later rejoining a class, is always complete.

Timetable (schedule)

The timetable is a list of weekly **slots**: a class-subject on a weekday from a start time to an end time, in a room. The system refuses a slot that clashes with another: the **same class** cannot be in two lessons at once, the **same teacher** cannot teach two lessons at once, and the **same room** cannot be used twice. The error lists exactly which lessons clash. Changing a subject's teacher is checked the same way.

Attendance

A teacher (or administrator) opens the attendance sheet for a lesson and date and sees the class roster. Each student is marked **present**, **absent**, **late** or **excused**, with optional remarks, and the whole sheet is saved at once. Saving again simply updates the marks, so nothing is duplicated.

- The roster is **dated**: it lists the students who were in the class on that day, so editing an older sheet shows who was really there.
- Future dates are refused, and so are dates outside the class's school year.
- **Attendance rate = (present + late) / all marked**. It is empty, not zero, when nothing has been marked yet.
- A student sees their own record and rate; a teacher sees summaries for their lessons; an administrator sees the school-wide figure for today.

Assessments and grades

An **assessment** is a quiz, test, exam, assignment or project for a class-subject, with a date, a term (1 to 3) and a **maximum score**. The teacher then enters a score for each student on the grade sheet. A score must be between 0 and the maximum, and the whole sheet is rejected if any row is invalid, so a bad entry never half-saves.

A student's result for a subject is **points-weighted**: total points scored divided by total points possible. A 20-point quiz counts more than a 5-point quiz, which is fairer than averaging percentages. Deleting an assessment also removes its grades, and the screen asks for confirmation and shows how many grades will go.

Announcements

An announcement has a title and body, an **audience** (everyone, teachers only, or students only), an optional **class** (otherwise it is school-wide) and optional **publish** and **expiry** dates. It appears only between those dates. Administrators can post to anyone. A teacher must pick a class they teach, and can edit or delete only their own posts. Students see only active announcements aimed at them or their class.

5. A day in the life

The administrator: setting up and running the school

- 1 **Create the people.** Users, Create user: teachers first, then students (or let students sign up themselves).
- 2 **Create the structure.** Subjects, then Classes for the new school year (name, grade level, homeroom teacher).
- 3 **Assign teachers.** Open a class, tab "Subjects & Teachers", and add each subject with its teacher.
- 4 **Enroll students.** Open a class, tab "Students", and enroll several at once, or enroll a single student from their page.
- 5 **Build the timetable.** Open a class, tab "Schedule", and add weekly slots. The system warns about any clash.
- 6 **Keep an eye on things.** The dashboard shows today's attendance, students without a class, upcoming assessments and recent announcements.
- 7 **Handle changes.** Transfer a student, change a teacher, deactivate a leaver, post an announcement.

The teacher: a normal teaching day

- 1 **Sign in** and read the "Today" dashboard: today's periods in order with the current one highlighted, plus pending grading.
- 2 **Mark attendance.** Pick the class and subject, tick present, late, absent or excused, and save.

- 3 **Create an assessment** (for example "Algebra Quiz 1", 20 points) and later open its **grade sheet** to enter every score.
- 4 **Post an announcement** to a class ("Bring your project on Friday").
- 5 **Check the timetable** for the week from the Schedule page.

The student: checking in

- 1 **Sign in** and see the dashboard: today's timetable, attendance rate and the latest grades.
- 2 **My class** shows the class, homeroom teacher and the subjects with their teachers.
- 3 **Attendance** shows the rate and a record of each day; **Grades** shows a percentage per subject and every graded item, with the next upcoming assessment.
- 4 **Announcements** lists what the school and the teachers have posted for them.

A student who has just signed up sees very little

Until an administrator enrolls them in a class, the student dashboard shows only their profile and a "not enrolled yet" message, because timetable, attendance and grades all belong to a class.

6. Rules the system enforces

These checks run in the API, so they apply no matter how a request is made. Where useful the screen also prevents the mistake early.

Area	Rule	What you see if you break it
Accounts	Email is unique. Role and email cannot change after creation.	A "already registered" message
Accounts	You cannot deactivate or delete yourself. The last active administrator is protected.	A clear refusal
Accounts	An account with school records can be deactivated but not deleted.	"Has school records, deactivate it instead"
Enrollment	One active enrollment per student. Closed enrollments are history and are kept.	"Already enrolled in a class"
Enrollment	Only current or future school-year classes can be enrolled into.	"Class is from a past academic year"
Classes	A class's school year and grade level cannot change once it has students or subjects.	"Class is in use"
Classes and subjects	Anything in use cannot be deleted.	"In use" (409)
Teachers	A teacher with current classes cannot be deactivated. Only an active teacher can be assigned.	A clear refusal
Timetable	No clash of class, teacher or room. End time after start time.	A list of the clashing lessons
Attendance	Only the assigned teacher (or an admin) can mark. No future dates, no dates outside the school year. Only students on that day's roster.	A "not allowed" or "date" message

Area	Rule	What you see if you break it
Grades	Only the assigned teacher (or an admin) can enter. Score between 0 and the maximum. The maximum cannot be lowered below a recorded score.	The row is rejected, nothing is saved
Announcements	Expiry must be after publish. Teachers can target only their own classes and edit only their own posts.	A "not allowed" message
Everything	Every request is checked for a real sign-in, an active account and the right role, even if the screen was bypassed.	Sign-in page, or a "no access" page

What the status numbers mean

- **401:** you are not signed in (or the session expired). The app asks you to sign in again.
- **403:** you are signed in but not allowed to do that. The app shows a "no access" page.
- **404:** that record does not exist. **409:** it conflicts with existing data (duplicate, in use, clash).
- **400:** the input is not valid. The form shows which field. **503:** a service the API depends on is briefly unavailable; the app retries quietly.

7. Dashboards and everyday features

Each role gets its own dashboard with different content, not the same page with different colours.

Role	First screen shows	Menu
Administrator	Counts of students, teachers, classes, subjects, active enrollments and students without a class (each opens the matching list); attendance today; enrollment by grade; upcoming assessments; recent announcements; buttons to create a user or an announcement.	Dashboard, Users, Students, Teachers, Subjects, Classes, Attendance, Grades, Announcements
Teacher	Today's periods in order with the current one marked; how many of today's attendance sheets are done; assessments still waiting for grades; their classes; recent announcements; a Mark attendance button.	Dashboard, My classes, Schedule, Attendance, Grades, Announcements
Student	Their identity card (class, school year, homeroom teacher); today's timetable; attendance rate; grades by subject; recent grades; upcoming assessments; announcements.	Dashboard, My class, Schedule, Attendance, Grades, Announcements

- **Search and filters.** Every list has a search box, filters (class, grade, status and so on), sortable columns and page controls. The choices are kept in the web address, so a filtered list can be bookmarked or shared.
- **Counts beside titles.** List pages show the running total ("Students 8"), and the page returns you to the last page if the one you were on empties.
- **Light and dark themes.** A switch in the top bar (and on the sign-in page) changes the whole look. The choice is remembered on the device, and your system setting is used until you pick one.
- **Phones.** On a phone the menu becomes a floating bar at the bottom with the four most-used pages and a "More" button for the rest.
- **Accessibility.** Everything works with a keyboard, focus is visible, pages have headings for screen readers, and colours meet contrast guidelines.

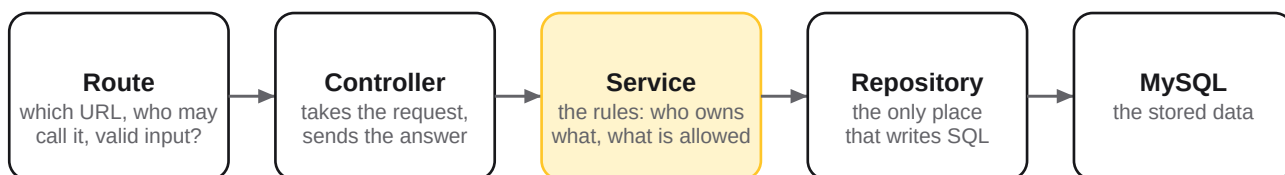
8. Under the hood

The technology

Layer	Built with	In plain words
Website	React 19, Vite, Tailwind CSS, TanStack Query	The screens, the styling, and a helper that fetches and caches data
API	Node.js 24, Express 5, zod	The back office and its input checks
Database	MySQL 8, 12 tables	Links between records are enforced by the database itself
Sign-in	Firebase Authentication	Email and password, reset emails, signed tokens
Documentation	OpenAPI and Swagger UI	A page listing all 64 API endpoints you can try out

How one request travels

Every feature uses the same four steps, in the same order, so a problem can be traced by following a single path.



- **Route:** matches the URL, requires a sign-in, names the roles allowed, and validates the input with strict rules (unknown fields are refused).
- **Controller:** a thin hand-off. It passes the checked input to the service and sends the result back in the standard format.
- **Service:** the business rules. It decides who owns what, applies the rules from section 6 and shapes the answer.
- **Repository:** the only place that writes SQL. Values are always passed as parameters, never glued into the text, which blocks SQL injection.

Every answer looks the same

Success: { "success": true, "data": ..., "meta": ... } where meta appears on lists (page, limit, total).

Failure: { "success": false, "error": { "code", "message", "details" } } with a code from a short fixed list. Each request also carries a request id that appears in the logs, so a user's problem can be matched to a log line.

Where the code lives

Folder	Contents
backend/src/modules/<feature>/	One folder per feature (users, students, classes, attendance, grades, and so on) with its routes, controller, service, repository and input rules
backend/src/modules/access/	The ownership rules every service calls: "is this teacher assigned to this class?", "is this student looking at themselves?"
backend/database/	schema.sql (the 12 tables and their constraints) and seed.sql (demo data)
backend/docs/openapi.yaml	The API contract shown at /api/docs
backend/tests/	128 automated tests that run against a real MySQL database
frontend/src/features/<feature>/	One folder per feature with its API calls, hooks, forms, components and pages
frontend/src/components/	Shared building blocks: buttons, tables, modals, the layout and navigation
DESIGN.md, docs/	The visual design system, the architecture overview and the project plan

How it stays safe

- Every request is verified: a real Firebase token, then an active account and a role looked up in MySQL.
- The API is the enforcement point. The website hides what you cannot do, but the API refuses it regardless.
- Strict input checking, parameterised SQL, and only the allowed websites may call the API from a browser.
- Registration is rate-limited, passwords never touch the school database, and secret keys are kept out of the code repository.
- The database enforces key rules itself (unique emails, one active enrollment, valid score ranges), so even a bug in the code cannot break them.

9. Hosting, data and limits

Where things are

What	Where	Cost	Good to know
Website	Firebase Hosting school-management-system-b0a2a.web.app	Free	Static files, always on
API	Render web service skole-api.onrender.com	Free	Sleeps after 15 idle minutes; the next request takes about a minute to wake it
Database	Aiven MySQL free plan	Free	1 GB; may be switched off after long inactivity (turn it back on in Aiven)
Sign-in	Firebase Authentication	Free	Needs no card

Where the data lives, and what happens when people change laptops

No school data is stored on anyone's computer. Teachers, students and administrators only use a browser, and everything is stored in the one cloud database. Signing in from a new laptop or phone shows exactly the same data. The browser keeps only two small conveniences: the signed-in session and the light or dark choice.

Is it "real time"?

Everyone shares the same data, and every save is written to the database immediately. If a teacher marks attendance, an administrator who opens that page a moment later sees it.

Open pages do **not** update by themselves. A screen treats what it loaded as fresh for 30 seconds and shows other people's changes the next time it loads that data (opening the page again, or reloading). Your own changes appear at once. If two people edit the same record, the last save wins. A true live feed (like a chat app) would need extra technology and was not part of the brief.

Running it on your own computer

- Install Node.js 24 and MySQL 8, create a free Firebase project, and copy the example settings files.
- Run the checks (`npm run doctor`), create the tables (`npm run db:migrate`) and demo data (`npm run db:seed`).
- Start the API and the website in two terminals (`npm run dev` in backend and in frontend) and open `http://localhost:5173`.
- The README has every step. The local setup works without Render, Aiven or Firebase Hosting.

The demo data

Account	Name	Role
admin@school.test	Amara Johnson	Administrator
admin2@school.test	Noah Bennett	Second administrator
teacher1@school.test	Alice Morgan	Teacher: Maths and Computer Science, homeroom of Grade 10 - A
teacher2@school.test, teacher3@school.test	Brian Chen, Carla Diaz	Teachers
student1@school.test ... student8@school.test	Daniel Okafor, Grace Kim and six more	Students in Grade 10 - A and Grade 10 - B

Password for every demo account: **Password123!**

10. Questions and glossary

Can there be only one administrator?

No. There can be as many as the school needs. The system only guarantees that at least one active administrator always remains.

Is the administrator above everyone?

Yes. The administrator manages accounts and all school data. Teachers and students cannot do any of that.

How does a teacher get an account?

An administrator creates it (Users, Create user, role Teacher) and gives the teacher the temporary password. Teachers cannot sign up themselves.

Can a student be promoted to teacher?

No. Roles are fixed. A mistaken account is deleted while unused and a new one is created.

Can a teacher see other teachers' classes?

Teachers can see class names and who teaches, but they can only mark attendance and enter grades for their own lessons.

What if a student moves to another class?

The administrator transfers them. The old enrollment is kept as history, and the new class takes over from that day.

Does a deactivated person lose their data?

No. Their records stay. They simply cannot sign in until an administrator reactivates the account.

Why did the page take a minute to open?

The free server sleeps when nobody has used it for 15 minutes, and wakes on the first request.

Glossary

Term	Meaning
Role	Administrator, teacher or student. Decides what an account may do. Fixed at creation.
Active / deactivated	Whether an account may sign in. Deactivating blocks access but keeps all records.
Class	A group of students for one school year, such as Grade 10 - A, 2026-2027.
Class-subject	A subject taught in a class by one teacher. The base of the timetable, attendance and grades.
Enrollment	A student's membership of a class, with dates. History is kept.
Roster	The list of students in a class on a given day.
Assessment	A quiz, test, exam, assignment or project with a maximum score.
Token	A signed proof of sign-in from Firebase that the app sends with each request.
API	The back office the website talks to. It checks permissions and reads and writes the database.
Endpoint	One address on the API that does one job, such as "list students".
Academic year	The school year, starting in August and written as 2026-2027.
Free tier	The no-cost plan of a hosting service, with limits such as sleeping when idle.